precisely

# Insurance Market Segmentation in Panama City by K-Prototypes Clustering

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# Our Team Members

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Source: Your paragrhttps://www.travelandleisure.com/most-beautiful-places-in-florida-8384751aph text

# Hurricane Milton

**$85 Billion**

in damages

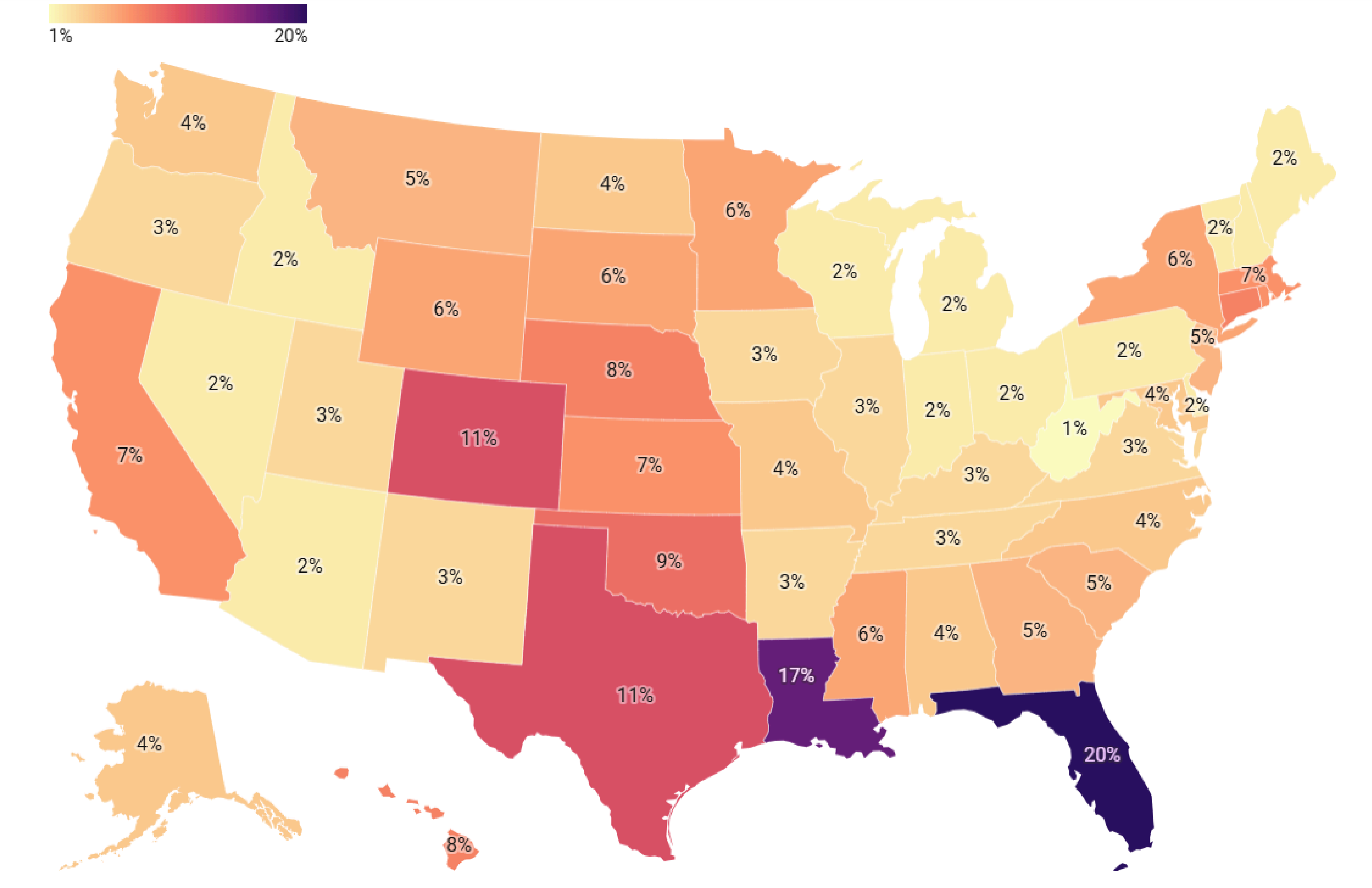
**285,000**

insurance claims

Source- https://www.axios.com/2024/10/11/hurricane-milton-florida-damage

Source- https://www.standard.co.uk/news/world/hurricane-milton-damage-photos-florida-tampa-bay-flooding-b1187150.html

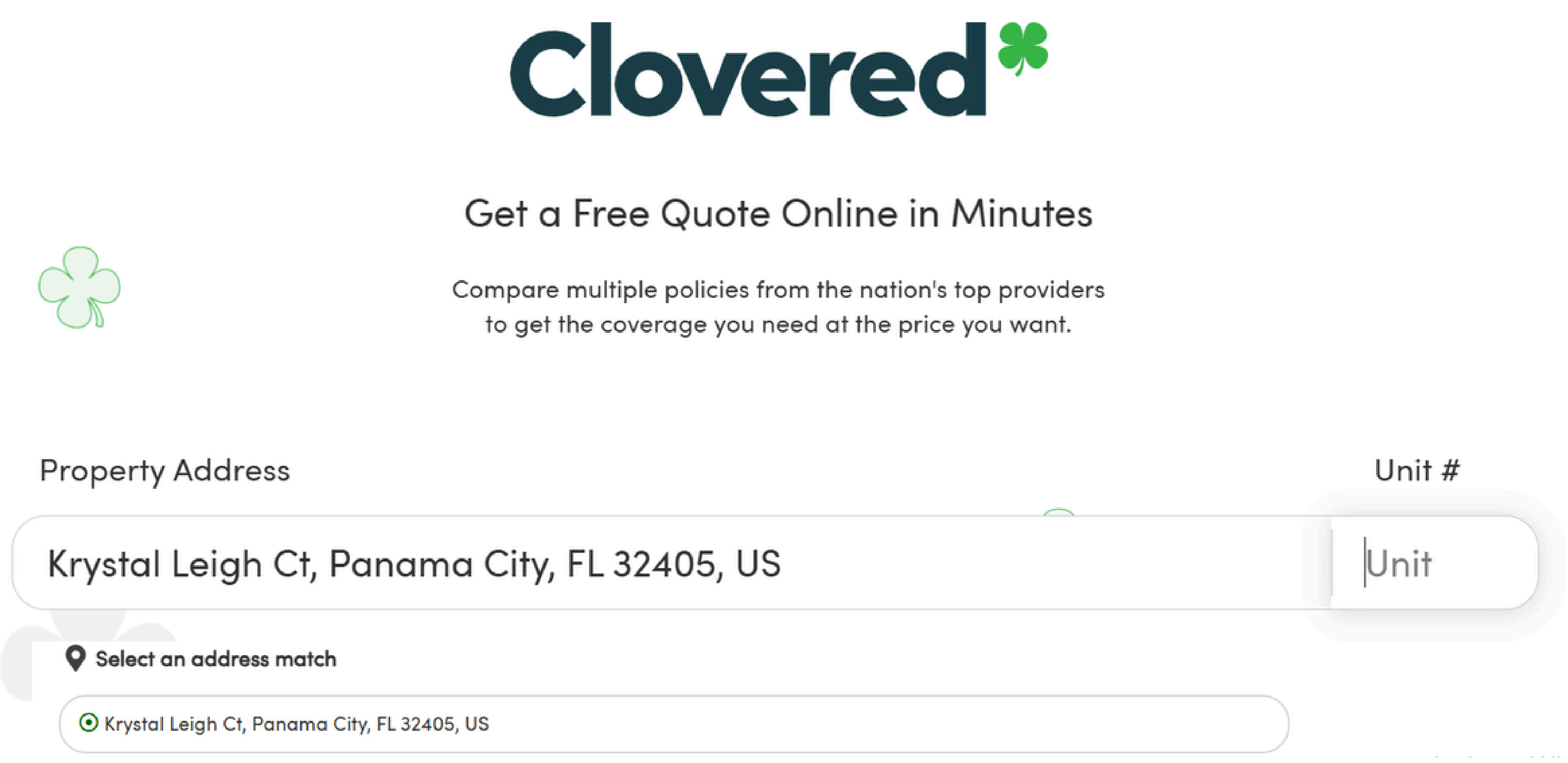
# Overview




- 20% of residential homeowners in Florida pay at least $4,000 per year, for home insurance.
- Average cost of flood insurance in **Panama City** is higher than other Florida counties.
- High Premiums are not the only solution.

Reference: https://www.realtor.com/news/trends/florida-home-insurance-cost-premiums/?msockid=2bee8033355964ae11799393343b65fa

# Market Requirements




- Occupancy
- Background Check
- Insurance History
- Applicant’s Age
- Household Income
- Style of Home
- Living Area (sqft)
- Age of Structure
- Building Material
- Flooring Material
- Roofing Material

Reference: https://quoting.clovered.com/Quote/

# Data Collection & Modeling

## Data Used

- Coastal Risk
- Flood Risk
- Demographic data
- Property Attributes

## Initial Approach

- Linear regression
- Most variables had minimum to no linear relationship

## New Approach

- New models used -
  Random Forest
  Gradient Boosting
- Selected 6 variables with highest significance out of 13 variables

# Flood Risk Score

$$\text{Flood Risk Score} = w1 \times \left(1 - \frac{\text{Elevation}}{\text{Max Elevation}}\right) + w2 \times \left(1 - \frac{\text{Floodzone Distance}}{\text{Max Floodzone Distance}}\right)$$

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$$w1 = \frac{\text{Importance Score of Elevation}}{\text{Importance Score of Elevation} + \text{Importance Score of Distance}} = 0.55$$

$$w2 = \frac{\text{Importance Score of Distance}}{\text{Importance Score of Elevation} + \text{Importance Score of Distance}} = 0.45$$

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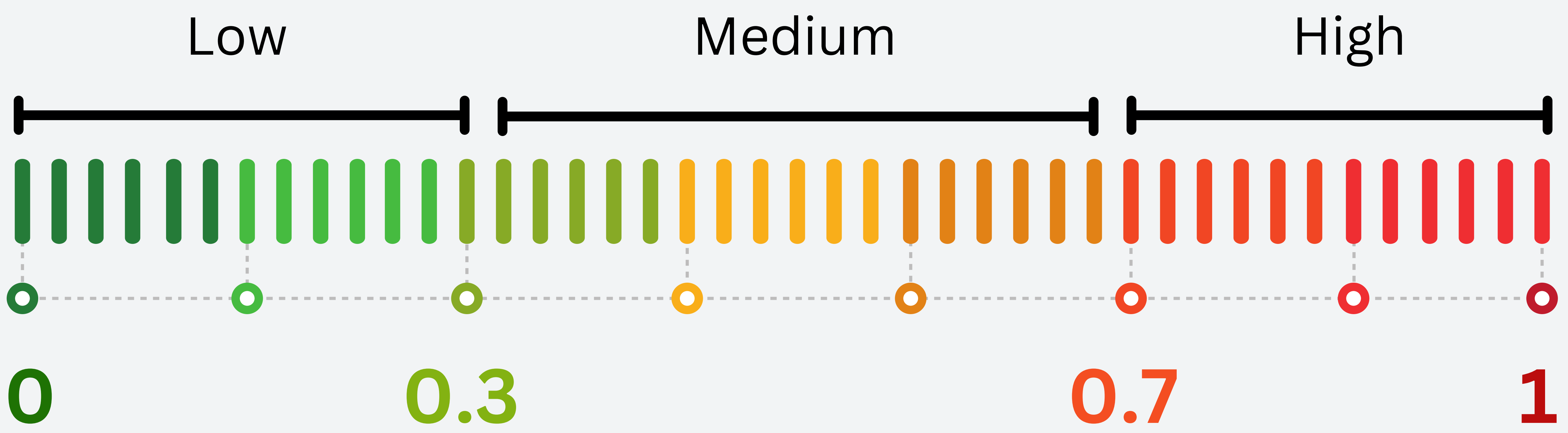


# Clustering Model

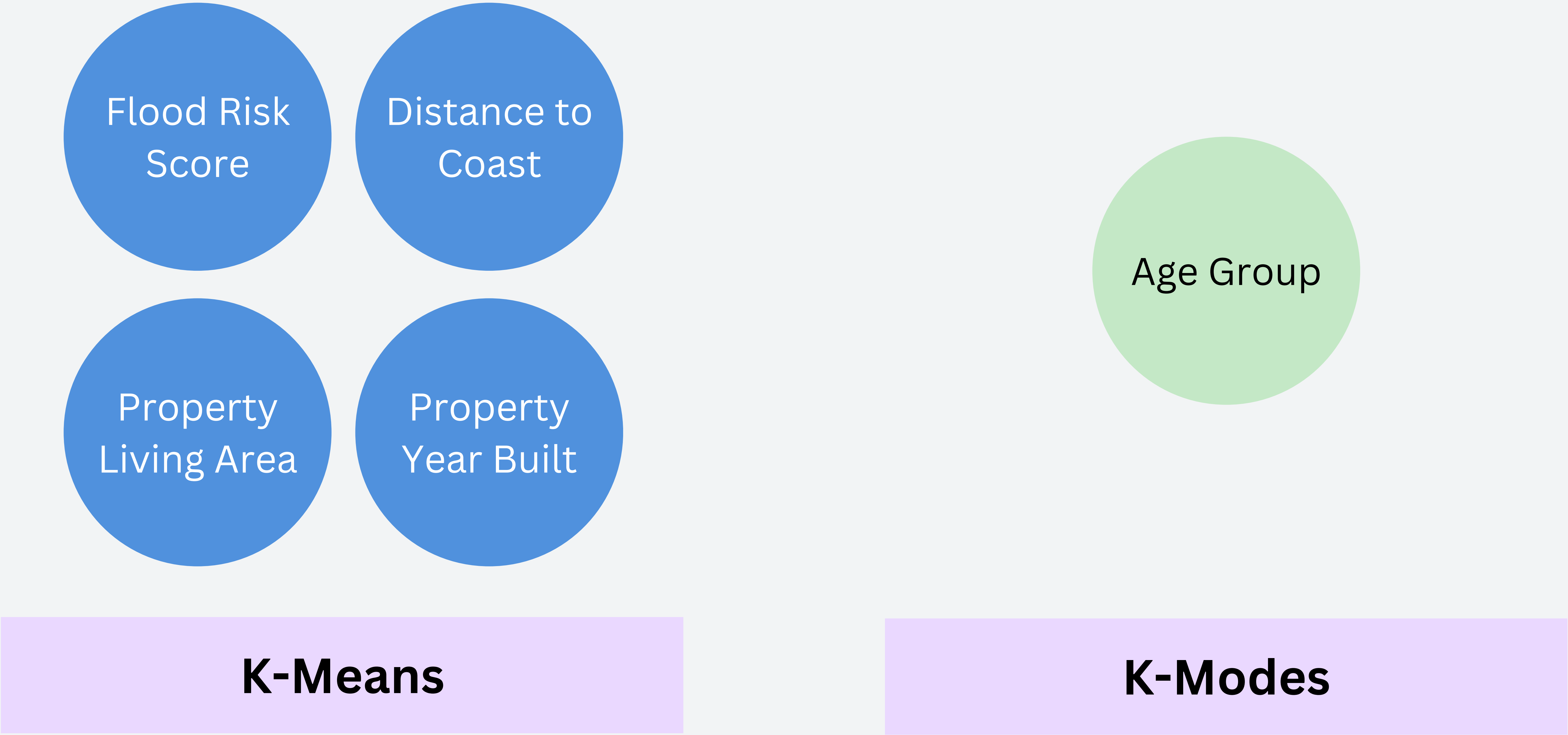
## Objectives:

**Segment Panama City's residential market** to inform exposure management based on flood and coastal risks, property characteristics, and demographics.

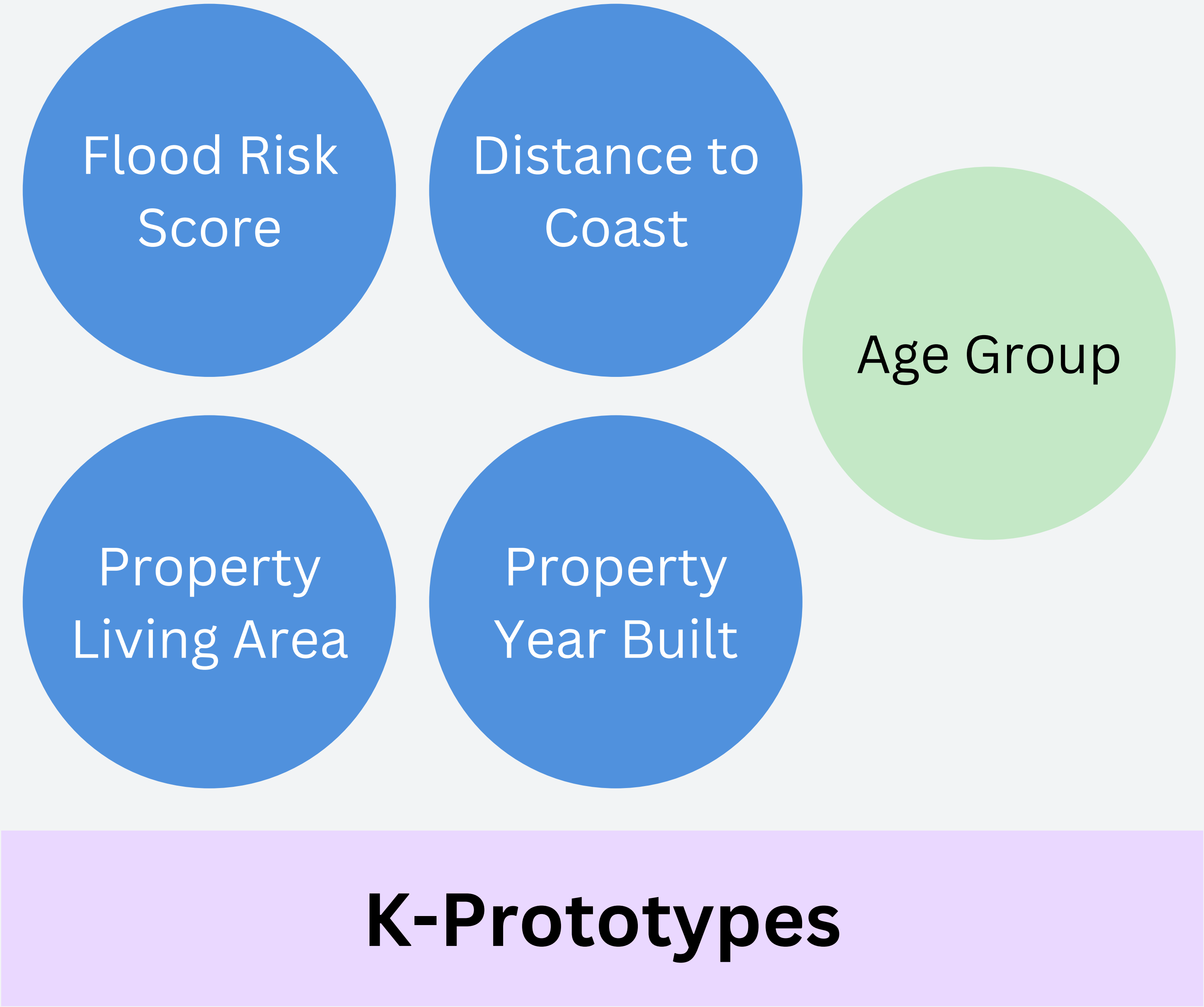
## Methodology:

- Cluster Identification: Used **Elbow Method** to determine **4 clusters**.
- Model: Applied K-Prototypes Clustering to handle mixed data.

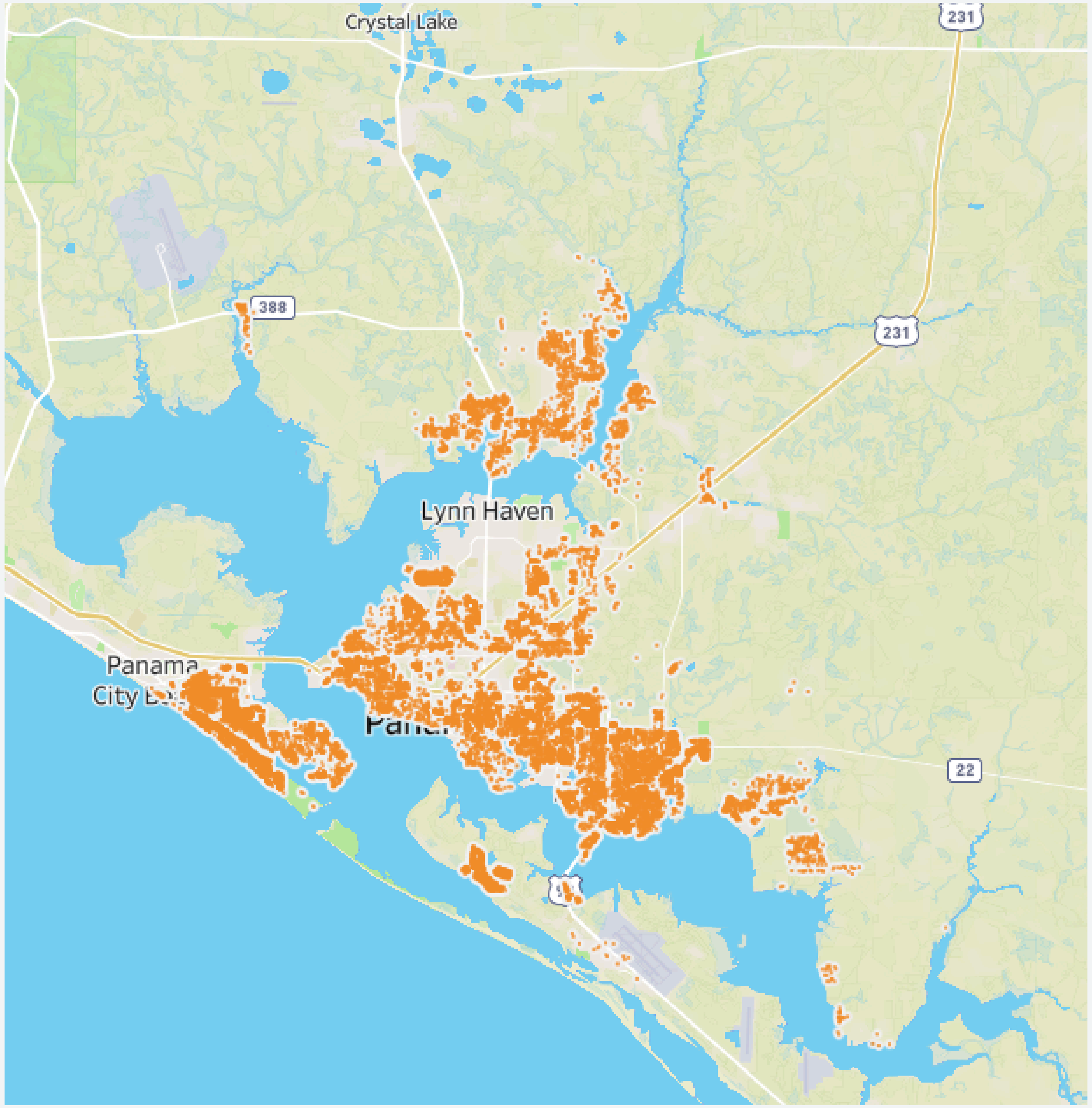
# Model Choice: K-Prototypes



# Model Choice: K-Prototypes




**Combines both K-Means and K-Modes**, making it suitable for our dataset which contains both numerical and categorical variables.




# Cluster 1

## High Flood Risk

Flood risk score: 0.85

## Small Properties

Living Area: 1399 sq ft

## Modern Properties

Year Built: 1995

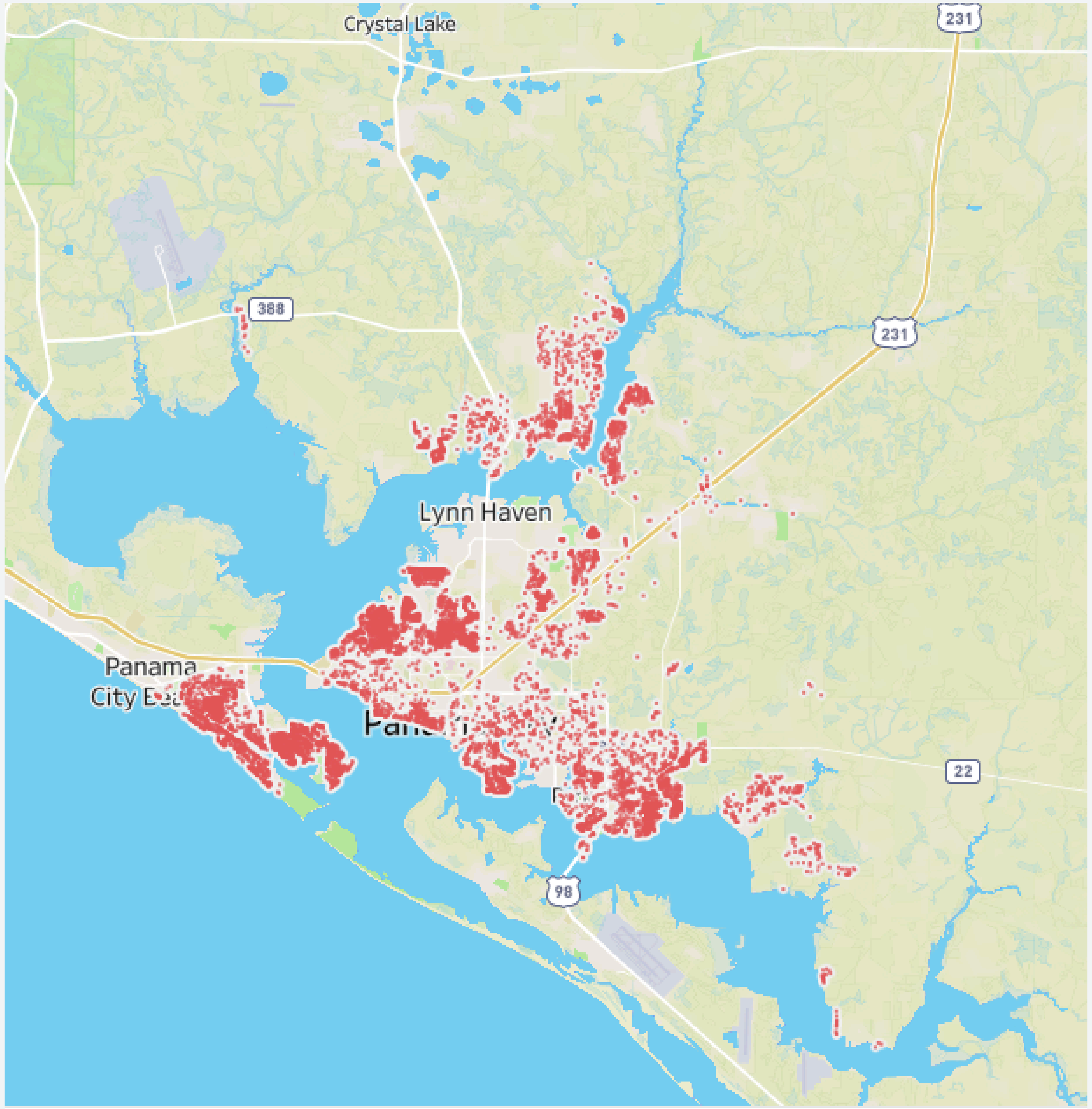
## Mixed-age Population

28.8% Above 50

37.9% Below 50

*These above statistics are the mean of the relative cluster*

# Cluster 2




## High Flood Risk

Flood risk score: 0.88

## Large Properties

Living Area: 2874 sq ft

## Modern Properties

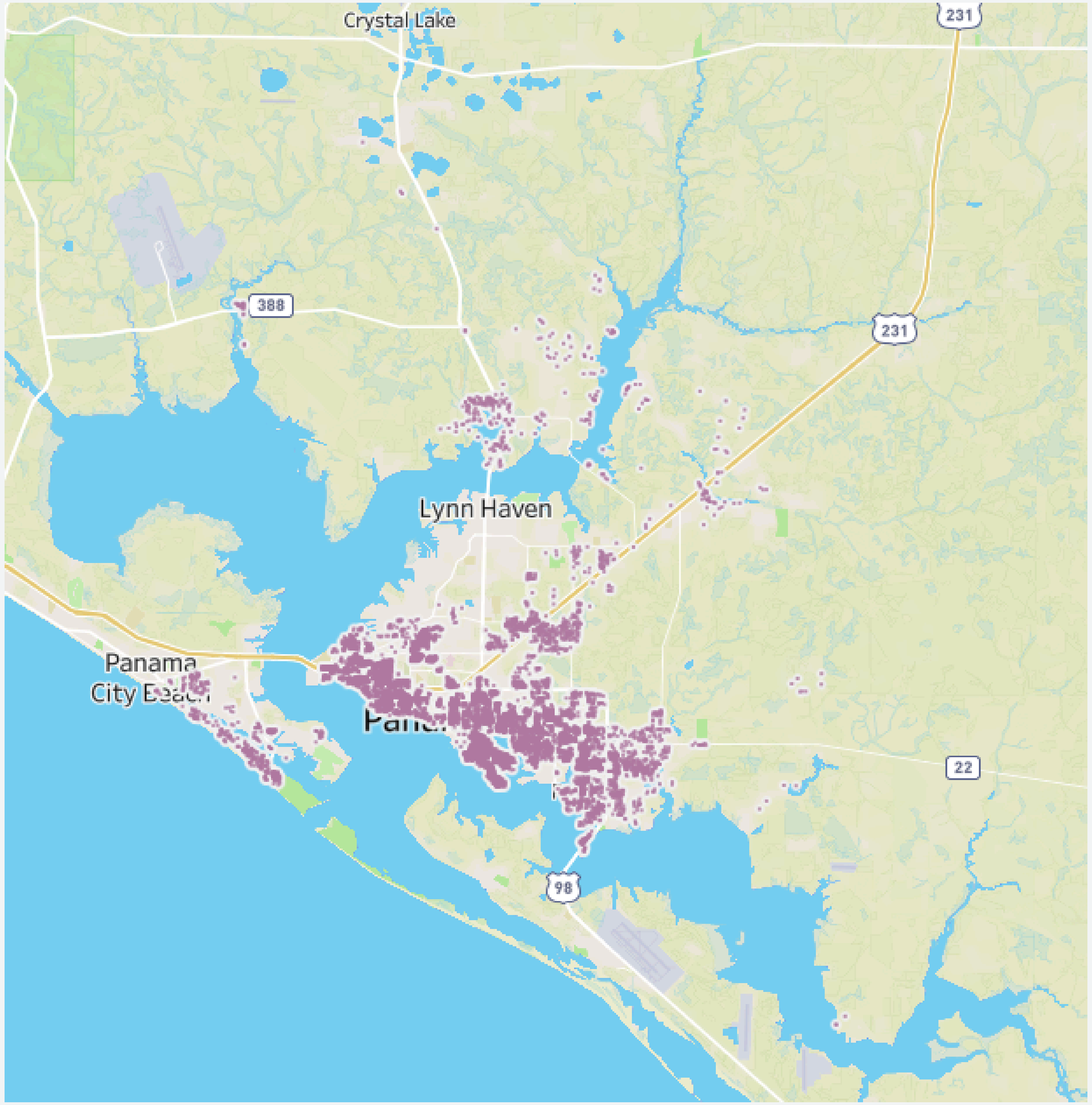
Year Built: 1993

## Older Population

51.1% Above 50

23.4% Below 50

*These above statistics are the mean of the relative cluster*




# Cluster 3

**Moderately High Flood Risk**

Flood risk score: 0.79

**Small Properties**

Living Area: 1260 sq ft

**Older Properties**

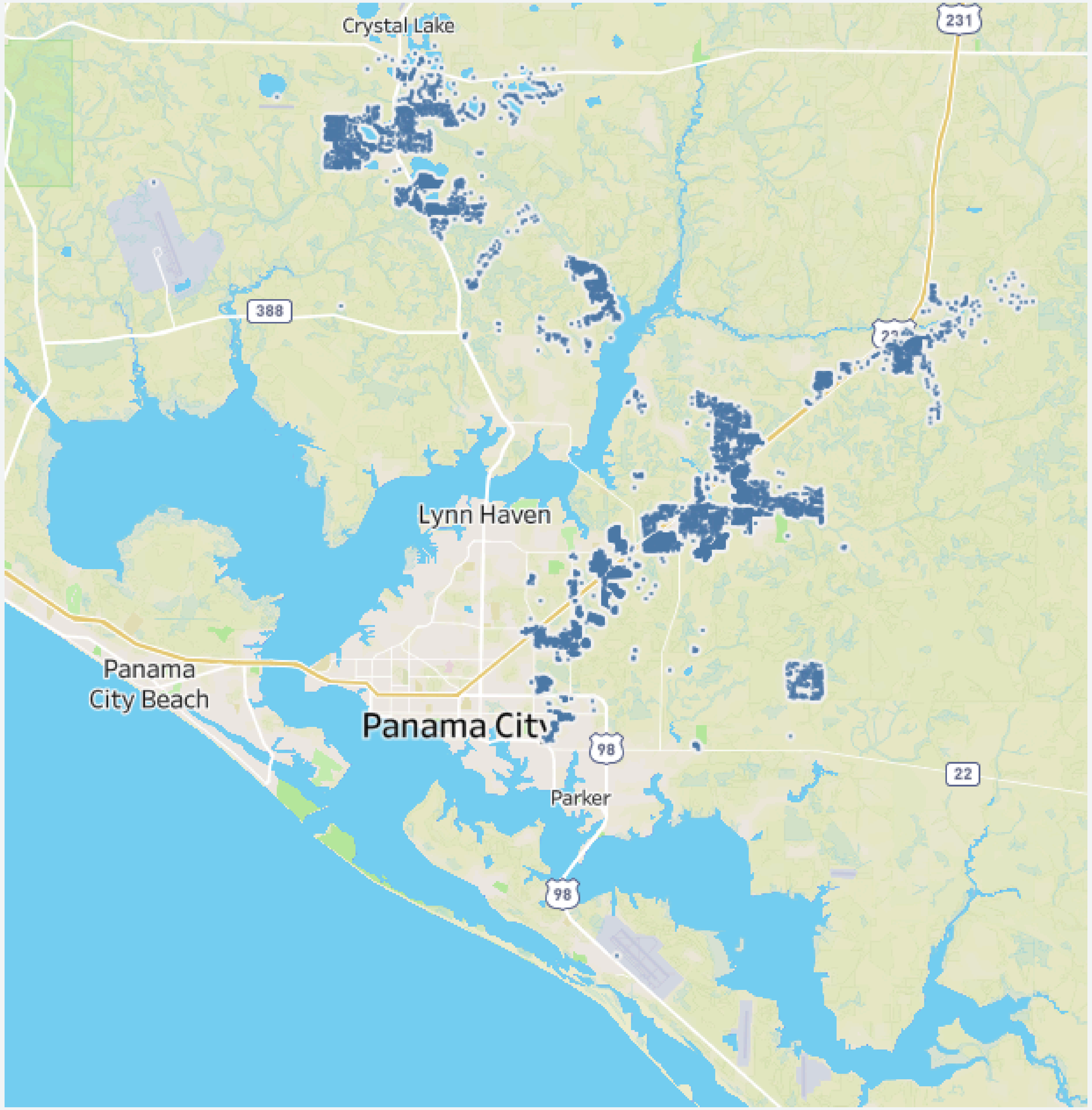
Year Built: 1954

**Mixed-age Population**

30.9% Above 50

29.3% Below 50

*These above statistics are the mean of the relative cluster*




# Cluster 4

## Moderate Flood Risk

Flood risk score: 0.66

## Medium Properties

Living Area: 1791 sq ft

## Newer Properties

Year Built: 2005

## Younger Population

19.9% Above 50

49.4% Below 50

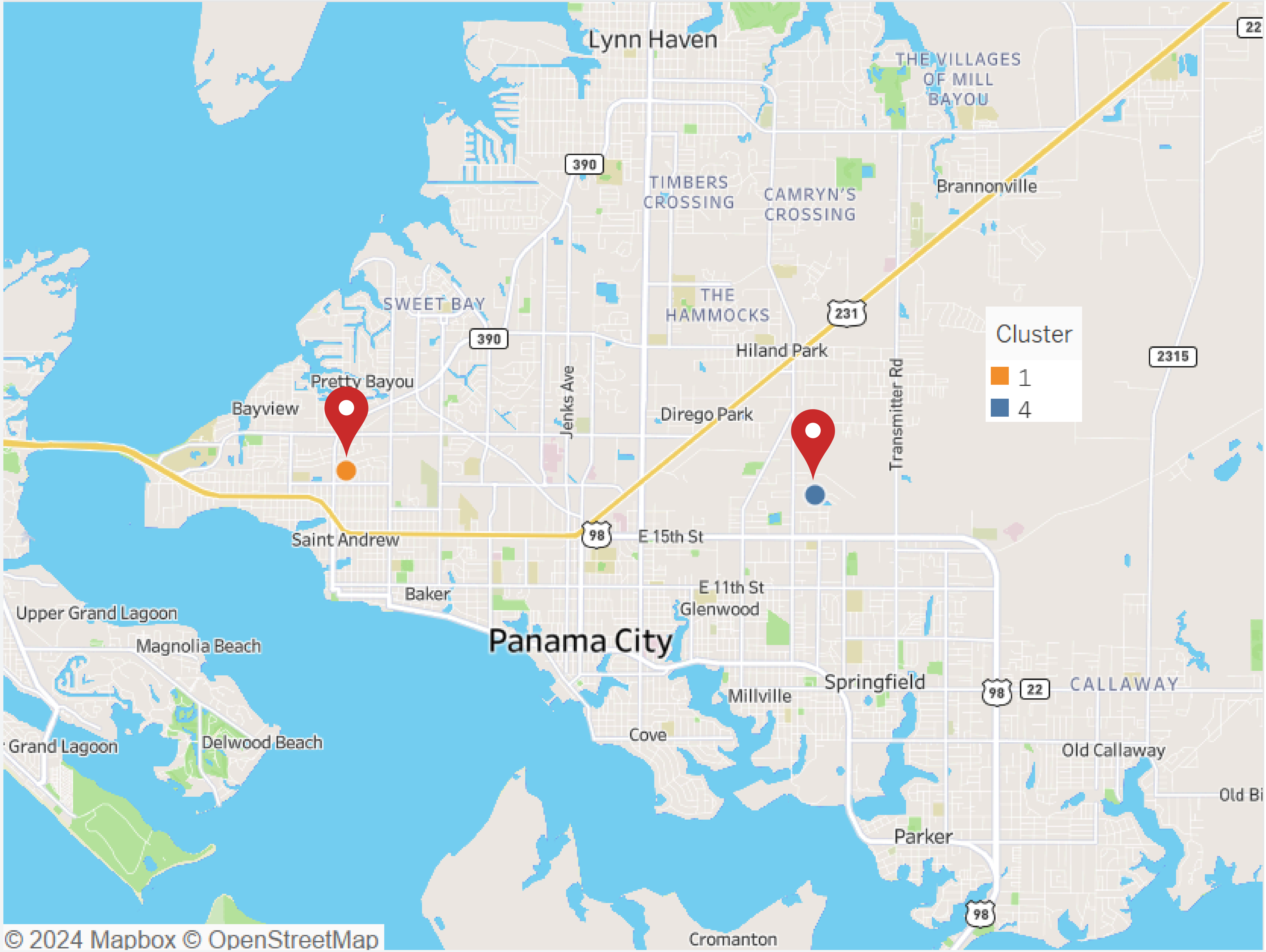
*These above statistics are the mean of the relative cluster*

- Year Built : 1967
- Living SqrFt : 1592
- Floodzone Distance : 0
- Dist to coast (Feet) : 2763
- Elevation (Feet) : 13
- Medium Income household

**Premium : $5,241**

- Year Built : 1998
- Living SqrFt : 2032
- Floodzone Distance: 1103
- Dist to Coast (Feet) : 13778
- Elevation (Feet) : 37
- Low Income household

**Premium : $11,231**


Lynn Haven
THE VILLAGES OF MILL BAYOU
390
TIMBERS CROSSING
CAMRYN'S CROSSING
Brannonville
SWEET BAY
390
THE HAMMOCKS
231
Hiland Park
Cluster
1
4
2315
Pretty Bayou
Bayview
Jenks Ave
Dirego Park
Transmitter Rd
98
E 15th St
Saint Andrew
E 11th St
Glenwood
Baker
Upper Grand Lagoon
Panama City
Magnolia Beach
Springfield
98
22
CALLAWAY
Millville
Cove
Grand Lagoon
Delwood Beach
Old Callaway
Parker
98
Cromanton
22
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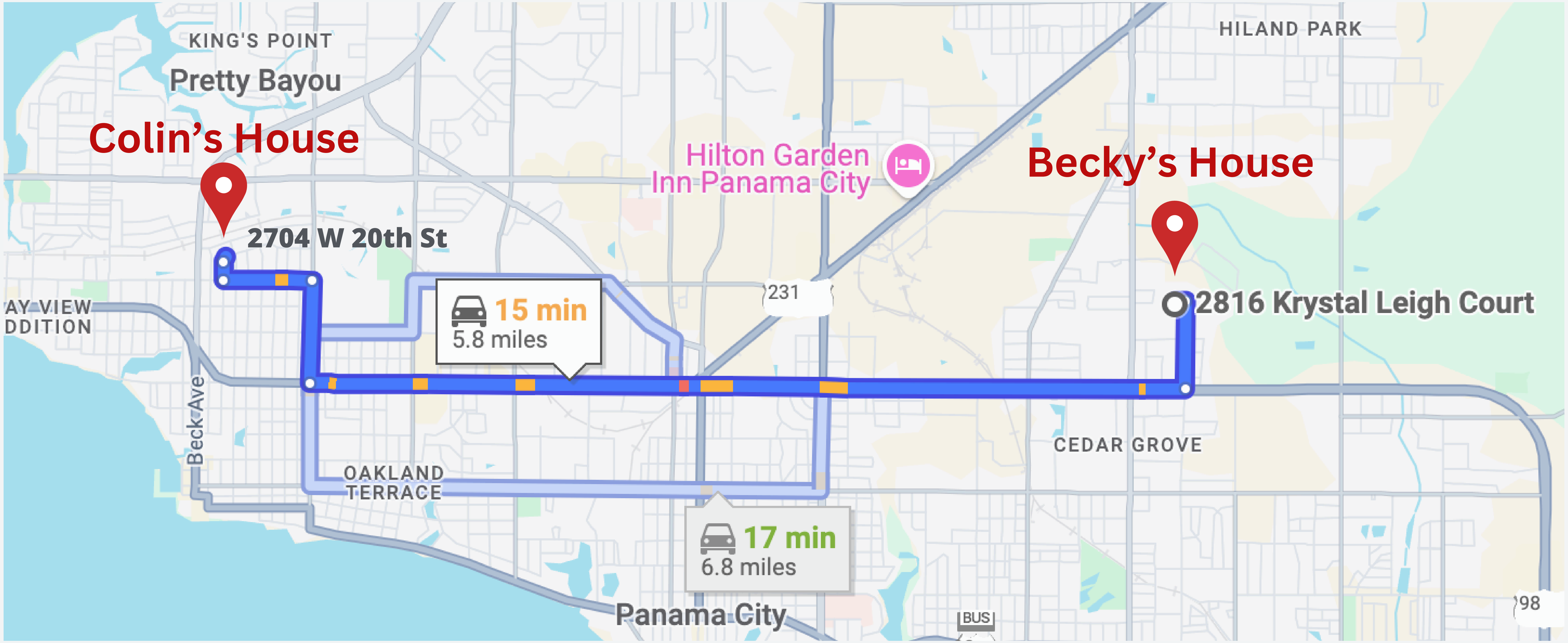

KING'S POINT
Pretty Bayou
Colin's House
2704 W 20th St
HILAND PARK
Hilton Garden Inn Panama City
Becky's House
2816 Krystal Leigh Court
231
15 min
5.8 miles
AY VIEW
DDITION
Beck Ave
OAKLAND TERRACE
CEDAR GROVE
17 min
6.8 miles
Panama City
BUS
98

# Recommendations

Cluster 1 + Cluster 2 = **60%** of the current portfolio

- Reassess portfolio with High-Risk Addresses
- Increase Risk Adjusted Premiums with Incentives
- Implement Catastrophic Loss Funds

**DECREASE EXPOSURE**

# Recommendations

Cluster 4
10%

- Expand Portfolio with Lower Risks Addresses
- Offer Competitive Premiums with Value-Added Services
- Promote Long-Term Policy Retention

INCREASE EXPOSURE

# Q&A

# Thank You